Predicting Used Car Value Retention Ratio Using Multi-Source Data and Machine Learning

1. Project description

This project predicts how much of its original sticker price a used car keeps, and finds which factors drive that loss. The prediction target is the **value retention ratio**:

$$\text{retention ratio} = \frac{\text{current listing price}}{\text{original base MSRP of that make, model and model year}}$$

A ratio of 0.60 means the car is listed at 60% of what the cheapest new version of that model year cost. The final study covers 3 makes with 3 models each: Toyota (Camry, Corolla, RAV4), Honda (Civic, Accord, CR-V) and Ford (F-150, Escape, Explorer), 1,908 listings in total. Nissan (Altima, Rogue, Sentra) was also collected but could not be modeled, as section 2 explains.

The pipeline has six steps, each a Python module in the repository:

1. **Collect** used-car listings from two independent sites, Craigslist (mostly private sellers) and CarMax (a national used-car retailer), plus MSRP data from U.S. News & World Report.
2. **Parse** each source into its own CSV, keeping the site's original column names and formats.
3. **Integrate** the sources into one schema, clean and de-duplicate, merge MSRP and compute the target.
4. **Engineer features** such as vehicle age, miles per year and keyword flags from the listing description.
5. **Train and compare** four regression algorithms with cross-validation and a held-out test set.
6. **Analyze depreciation** with permutation importance, regression coefficients, a log-linear depreciation model and partial-dependence curves.

2. Data collection

Three web sources were scraped on October 23, 2026: 1,312 Craigslist listings and 1,560 CarMax listings for 12 models, plus original MSRP ranges for 208 model years from U.S. News. Both listing sources clear the 500-listing minimum on their own, before any integration.

Source	Type of seller	What was scraped	Records	Columns
Craigslist, cars + trucks within 150 miles of Philadelphia (ZIP 19103)	Private owners (47% of kept listings)	The search-results page for each make/model, then every listing's detail page: attribute table, posting text, map location, post dates	1,312 (up to 110 per model)	33

Source	Type of seller	What was scraped	Records	Columns
	and small dealers			
CarMax, nationwide inventory ranked nearest-store-first for ZIP 19103	One national used-car retailer	The JSON endpoint that fills CarMax's search page, 100 cars per request	1,560 (130 per model)	46
U.S. News & World Report cars	MSRP reference, not listings	Each model-year review page (2000-2027 requested); the "Original MSRP" range from the cheapest to the most expensive trim	208 make/model/year ranges (2007-2027)	9

Makes and models. The models are high-volume ones that both sites carry in large numbers: Toyota Camry, Corolla, RAV4; Honda Civic, Accord, CR-V; Ford F-150, Escape, Explorer; Nissan Altima, Rogue, Sentra. All 12 were collected; Nissan was later left out of modeling (see the challenges below), so the final study has 3 makes with 3 models each.

Attributes collected. Craigslist gives title, price, model year, make/model, odometer, condition, cylinders, drive, fuel, paint color, title status, transmission, body type, VIN (when the seller adds it), seller type (owner or dealer), area, latitude/longitude, post and update dates, photo count and the seller's free-text description. CarMax gives stock number, VIN, year, make, model, trim, body, price, mileage, exterior and interior color, transmission, drivetrain, engine size and type, cylinders, horsepower, torque, city/highway MPG, store and state, distance from the ZIP, rating, a feature list, highlight badges (for example "single owner"), packages, prior use (rental, lease, loaner) and the date the car went on sale.

How the scrapers work. Each collector is a JavaScript file in `scrapers/` that runs in a browser tab on the site itself and parses pages with the browser's `DOMParser`. Running inside the site keeps every request identical to a normal page load (same cookies and headers) and let the CarMax collector read the same JSON its search page uses. Requests were spaced out: 2 parallel workers with 450 ms pauses on Craigslist, 800 ms on CarMax and 400 ms on U.S. News. Each collector saves a JSON file, and `parse_raw.py` turns each one into a CSV that keeps the site's own column names and formats (`data/raw/craigslist_listings_parsed.csv`, `data/raw/carmax_listings_parsed.csv`, `data/raw/usnews_msrp_parsed.csv`).

Discussion and challenges.

- **Sites that could not be used.** Cars.com answered with a Cloudflare "verify you are human" page, and Facebook Marketplace requires a login. Neither check was bypassed; CarMax was used as the second source instead.
- **Craigslist volume.** A single city has only about 60 listings of a given model, so the search used a 150-mile radius around Philadelphia, which pulls in New York, New Jersey, Delaware, Maryland and eastern Pennsylvania areas. Wider searches also bring cross-posted duplicates, which the search's duplicate bundling and the cleaning step remove.
- **Rate limiting at U.S. News.** After about 210 model-year pages at a 400 ms delay, U.S. News stopped answering requests from my browser for the rest of the session. The pages for the Nissan Rogue (2015-2026) and every Sentra year were never read. Without an MSRP the target cannot be computed, and Nissan would have kept only one complete model, below the 3-models-per-make rule, so Nissan was dropped from modeling. The Nissan rows are still in the raw files and the cleaning log. The collector now notes that a delay of 2 seconds or more should be used.
- **MSRP coverage.** U.S. News has model-year pages from 2007 onward, so older Craigslist cars cannot get a target and are dropped during integration.
- **Messy seller input.** Some Craigslist prices are placeholders ($1, $123) or down payments ("$2,000 down"), and some year fields disagree with the title or VIN. These are handled in preprocessing.

3. Preprocessing and integration

Both sources were mapped onto one schema, cleaned by logged rules and joined to the MSRP table, leaving **1,908 listings** (744 Craigslist, 1,164 CarMax) with 24 listing attributes from the sites, 2 MSRP values and the target. `integrate.py` does every step and writes the per-rule counts to `results/cleaning_steps.csv`.

Resolving names, formats and units

Unified attribute	Craigslist field and format	CarMax field and format	How they were unified
price (USD)	`price` text, "$13,425"	`basePrice` number, 13425.0	strip "$" and commas, convert to number
mileage (miles)	`attr_odometer` text, "116,700"	`mileage` number	strip commas, convert to number
year	`year` attribute, year in the title, VIN	`year` number	VIN 10th character first, then the title year, then the attribute; 7 listings corrected

Unified attribute	Craigslist field and format	CarMax field and format	How they were unified
make, model	make_model free text ("toyota rav4 high")	make, model ("RAV4 Hybrid")	regular expressions per model; other models dropped (Corolla Cross, Rogue Sport, Explorer Sport Trac)
trim	words in the title	trim ("XLE")	matched against a 39-name trim vocabulary; "Base/Unknown" if none
body_type	attr_type: sedan, SUV, pickup, truck	body: "4D Sport Utility", "Crew Cab Pickup"	7 classes: sedan, suv, truck, hatchback, coupe, wagon, other
drivetrain	attr_drive: fwd, rwd, 4wd	driveTrain: "All Wheel Drive"	fwd, rwd, awd/4wd (Craigslist does not separate AWD from 4WD)
transmission	automatic, manual, other	"Automatic"	automatic or manual ("other" on Craigslist is CVT or automated)
fuel_type	attr_fuel	engineType; "Hybrid" in the model name	gas, hybrid, plug-in hybrid, electric, diesel
cylinders	"4 cylinders"	4	number
exterior_color	attr_paint_color	exteriorColor ("Magnetic Gray")	6 color groups + other + unknown
condition	excellent, good, like new, fair, salvage	not published	CarMax rows = "not stated"
title_status	clean, rebuilt, salvage, lien	not published	CarMax rows = "clean" (CarMax does not retail salvage or flood titles)
seller_type	breadcrumb "by owner" / "by dealer"	always a dealer	owner or dealer
market	Craigslist area ("north jersey")	store city and state	lower-case label (130 values)
distance_miles	from latitude/longitude to	distance field from ZIP 19103	miles

Unified attribute	Craigslist field and format	CarMax field and format	How they were unified
	Philadelphia (haversine)		
listed_date	posted_at	lastMadeSaleableDate	date
description	seller's posting text	highlight badges, prior-use notes, feature list and packages joined into sentences	text
msrp_base, msrp_top			U.S. News range "$22,970 - 31,370" split into two numbers, joined on make + model + year

Cleaning and de-duplication

Rule	Craigslist removed	CarMax removed
Nissan, outside the final scope	322	390
Not the searched model, or a "we buy cars" ad	20	0
Same listing ID or same VIN	0	0
Repost: same model, year, price and mileage	21	0
Price missing or outside $1,000-$120,000	83	0
Odometer missing or implausible (under 100 miles on a car 2+ years old, over 400,000)	1	0
Listed price is a down payment ("$2,000 down")	9	0
No MSRP for that make/model/year (model years before 2007)	112	0
Retention ratio outside 0.03-2.0	0	6
Remaining	**744**	**1,164**

No VIN appeared in both sources, so the two sites did not overlap. Duplicates came from Craigslist sellers reposting the same car; only 45% of Craigslist listings include a VIN, so reposts were also matched on model, year, price and mileage.

Missing values. In the raw Craigslist data the VIN is missing in 56% of listings, condition in 33%, paint color in 30%, cylinders in 27%, drive in 26% and price in 5%; CarMax fields are nearly complete. Body type, drivetrain, transmission, fuel and cylinders were filled with the most common value for that model, because those depend mostly on the model. Condition and color were kept as their own "not stated" and "unknown" categories, since a missing value can itself carry information (dealers often skip condition).

Attribute statistics

Attribute (median, range)	Craigslist (744)	CarMax (1,164)
Price	$9,000 ($1,000-$41,995)	$23,998 ($10,998-$65,998)
Mileage	122,483 (100-338,530)	48,677 (116-127,908)
Model year	2014 (2007-2026)	2020 (2014-2027)
Base MSRP	$23,070	$24,885
Retention ratio	0.41	0.99
Seller	347 owners, 397 dealers	1,164 dealer

Across all 1,908 listings: body type is 863 SUV, 763 sedan and 202 truck; drivetrain is 970 AWD/4WD, 898 FWD and 40 RWD; 1,886 are automatic; 1,738 run on gas and 167 are hybrids; 1,855 have a clean title, 27 rebuilt and 9 salvage. Every model has 199-226 listings. The two sources describe different markets: CarMax sells newer, low-mileage cars at retail prices, while Craigslist is mostly older, high-mileage cars sold privately or by small lots. Retention ratios above 1.0 are common at CarMax (48% of its listings) because the denominator is the cheapest trim's price when new, while CarMax stock leans to mid and upper trims, and used-car prices are still inflated after 2021.

4. Feature engineering

The models get 29 inputs: 16 numeric features (3 as collected, 7 derived, 6 read from descriptions), 11 low-cardinality categories and 2 high-cardinality categories. `features.py` builds them and `train.py` encodes them inside each model pipeline.

Numeric features

Feature	Definition	Why it may matter
age_years	2026 - model year (0 for 2026-2027 cars)	the main clock of depreciation
mileage	odometer reading	wear and remaining life

Feature	Definition	Why it may matter
miles_per_year	mileage / age (age floored at 0.5)	separates hard-driven cars from garage cars of the same age
msrp_base	cheapest trim's MSRP for that model year	price level of the model
msrp_spread	top-trim MSRP / base MSRP	how much room there is for an expensive trim (median 2.3x for the F-150, 1.3x for the Corolla)
cylinders	engine cylinders	engine size
distance_miles	miles from Philadelphia	regional price differences
days_listed	scrape date - date posted / put on sale	stale listings may be overpriced
desc_word_count	words in the description	effort the seller put in
premium_feature_count	how many of 9 premium-feature groups the text mentions (leather, sunroof, navigation, heated seats, backup camera, CarPlay/Android Auto, blind spot, remote start, premium audio)	better-equipped cars

Features extracted from listing descriptions

Six yes/no flags come from regular expressions run over the description and title. For Craigslist the text is the seller's own; for CarMax it is the listing's badges, prior-use notes and feature list. The table compares listings with and without each phrase inside the same source, because comparing across sources would mostly measure the Craigslist-vs-CarMax gap.

Flag	Matches, for example	Share of Craigslist / CarMax listings	Median ratio with vs without
desc_one_owner	"one owner", "1-owner", "single owner"	13% / 58%	Craigslist 0.50 vs 0.40; CarMax 1.02 vs 0.95
desc_no_accidents	"no accidents", "clean Carfax", "accident free"	16% / 0%	Craigslist 0.49 vs 0.40

Flag	Matches, for example	Share of Craigslist / CarMax listings	Median ratio with vs without
desc_needs_work	"needs work", "mechanic's special", "as is", "check engine"	7% / 0%	Craigslist 0.24 vs 0.42
desc_recent_service	"new tires", "just serviced", "service records"	26% / 0.2%	Craigslist 0.41 vs 0.41
desc_warranty	"warranty"	11% / 23%	Craigslist 0.64 vs 0.38; CarMax 1.11 vs 0.95
desc_prior_rental_fleet	"rental", "fleet", "loaner", "previously leased"	1% / 50%	CarMax 0.98 vs 1.00

The flags behave as expected: "needs work" cuts the median Craigslist ratio by about 40%, while one owner, no accidents and warranty go with higher prices. "Recent service" shows no difference, likely because sellers of older cars mention repairs most often. Many of these signals overlap with age and mileage, which limits how much they add to a model (section 7).

Encoding the high-cardinality features

Two categories have many levels: **model_trim** (model + trim, 79 values, for example "F-150 Lariat") and **market** (Craigslist area or CarMax store, 130 values, many with 1-5 listings). Low-cardinality categories (source, make, model, body type and 7 others) are one-hot encoded. For the two high-cardinality columns, four options were compared with 5-fold cross-validation on the training set only:

Encoding of model_trim and market	Input columns	Ridge CV R^2	Ridge CV MAE	Gradient boosting CV R^2	Gradient boosting CV MAE
Dropped (baseline)	65	0.906	0.076	0.934	0.060
Frequency encoding (share of training rows)	67	0.908	0.074	0.941	0.058
Target encoding (mean ratio per level, 5-fold cross-fitted, smoothed)	67	0.910	0.074	0.937	0.059
One-hot, levels with fewer than 3 rows pooled into one "infrequent" column	215	**0.927**	**0.067**	**0.943**	**0.056**

All three encodings beat dropping the columns, so trim and market carry information. Target encoding packs everything into one column per feature and is the usual choice for high

cardinality, but here it lost to one-hot. A trim's average ratio mixes new and old cars of that trim, so the single number is blurred by age; separate one-hot columns let the model learn a trim's effect on top of age and mileage. Pooling rare levels keeps one-hot from creating columns for markets seen once. With fewer than 2,000 rows and only about 200 levels in total, the extra columns are affordable. The pipeline picks the encoding with the best mean CV R^2 automatically, so all four final models use pooled one-hot encoding; target encoding stays in the code as an option.

5. Model development

Four regression algorithms were trained on the same 29 inputs, each tuned by 5-fold cross-validation on the training set and scored once on a held-out test set.

Split. 80% train (1,526 listings) and 20% test (382), stratified by model so all 9 models appear in both parts in the same proportions (`random_state = 42`). The test set is used only once, after tuning.

Pipelines. Every model is a scikit-learn `Pipeline`, so imputation, scaling and encoding are fit on the training folds only and never see validation or test rows. Numeric features are median-imputed; for Ridge and k-NN they are also standardized, since both depend on feature scale. Categories are one-hot encoded as described in section 4.

Tuning and metrics. `GridSearchCV` with shuffled 5-fold `KFold` scores every setting on MAE, RMSE and R^2, and refits the setting with the lowest cross-validated MAE. A mean-ratio baseline (predict the training average for every car) shows what the models add.

Algorithm	Why it was included	Grid searched	Chosen setting
Ridge regression	linear baseline whose coefficients can be read directly	alpha: 0.1, 1, 10, 30	alpha = 1
k-nearest neighbors	prices of similar cars ("comps"), no fitted formula	k: 5, 10, 20; uniform or distance weights	k = 10, distance-weighted
Random forest (300 trees)	averages deep trees; handles interactions and non-linear depreciation	min leaf size: 1, 3; features per split: 33%, 60%	leaf 1, 60%
Gradient boosting (learning rate 0.05, 80% subsampling)	builds trees in sequence, each fixing the last one's errors	trees: 300, 600; depth: 3, 4	600 trees, depth 4

The target is the ratio itself, so an error of 0.05 means the prediction is off by 5% of the base MSRP. To make that concrete, test errors are also converted to dollars by multiplying the predicted ratio by the car's base MSRP and comparing with its listing price.

6. Evaluation results

Gradient boosting is the best model: test R^2 0.935 and test MAE 0.055, an average miss of about $1,341 per car. Its cross-validated and test scores agree closely, so it is not overfitting the training set.

Model	CV MAE	CV RMSE	CV R^2	Test MAE	Test RMSE	Test R^2
Gradient boosting	**0.052 ± 0.003**	**0.076 ± 0.008**	**0.949 ± 0.010**	**0.055**	**0.088**	**0.935**
Random forest	0.062 ± 0.004	0.089 ± 0.008	0.930 ± 0.010	0.067	0.106	0.907
Ridge regression	0.067 ± 0.003	0.091 ± 0.005	0.927 ± 0.008	0.073	0.106	0.905
k-nearest neighbors	0.092 ± 0.004	0.124 ± 0.007	0.863 ± 0.015	0.090	0.127	0.865
Mean-ratio baseline				0.289	0.345	0.000

CV values are the mean ± standard deviation over the 5 folds of the training set; the target is the retention ratio, so MAE 0.055 means 5.5% of the base MSRP.

Discussion.

- **Every model beats the baseline by a wide margin.** The baseline's MAE of 0.289 is about five times the best model's, so the features explain most of the spread in retention.
- **Tree ensembles win, but the linear model is close.** Ridge regression reaches test R^2 0.905, almost tied with the random forest, because depreciation is fairly smooth in age and mileage. Gradient boosting adds about 0.03 of R^2 by capturing interactions, for example that the gap between CarMax and Craigslist prices widens as cars age and that F-150 trims spread much more than Corolla trims.
- **k-NN is weakest.** With 200+ one-hot columns, distances are dominated by category mismatches rather than age and mileage, so its "nearest" cars are often not comparable.
- **Where the errors are.** For gradient boosting, the test MAE is 0.040 on CarMax listings and 0.077 on Craigslist listings: private-sale prices depend on seller urgency and condition details the data does not capture. By model, the RAV4, Accord, CR-V and Corolla are easiest (MAE about 0.04) and the F-150 (0.084) and Civic (0.097) hardest, because both have trims that change the price a lot (Raptor and Platinum for the F-150, Si and Type R for the Civic) and many Craigslist titles do not name the trim.
- **In price terms.** Converted to dollars, the median gradient-boosting miss is 4.6% of the listing price and 73% of test cars are predicted within 10% of their price.

7. Depreciation analysis and feature ranking

Where a car is sold, how far it has been driven and how old it is decide most of its retention. Holding the other factors fixed, a car keeps about 2.8% less of its base MSRP for each extra year and 3.7% less for each extra 10,000 miles; Toyota and Honda hold value much longer than Ford.

Feature ranking

Four methods were used: permutation importance on the test set (how much test R^2 drops when one feature is shuffled) for three models, and the tree-split (impurity) importance of gradient boosting. One-hot columns are added back to their original feature, so each feature is ranked once.

Feature	Gradient boosting, permutation	Random forest, permutation	Ridge, permutation	Gradient boosting, impurity
Source (Craigslist vs CarMax)	1	1	3	1
Mileage	2	2	2	2
Vehicle age	3	4	1	4
Model	4	3	5	3
Premium features mentioned	5	5	10	7
Make	6	6	6	8
MSRP spread (top/base trim)	7	10	9	6
Base MSRP	8	8	4	11
Model + trim	9	11	7	5

The same five features lead under every method: source, mileage, age, model and equipment. The models disagree on age versus mileage because the two move together (Spearman correlation 0.74): the linear model credits age, the tree models credit mileage. The six description flags rank low (each moves gradient boosting's test R^2 by less than 0.002) even though section 4 shows real price gaps, because what they say ("needs work", "one owner") is mostly already carried by mileage, age and source.

Direction of the effects (Ridge coefficients)

Ridge coefficients give the sign. On standardized features, one standard deviation of age (4.8 years) lowers the ratio by 0.136 and one standard deviation of mileage (56,000 miles) by 0.106. Being a CarMax listing adds 0.085 relative to the average and a Craigslist listing subtracts the same. The largest trim effects are on the F-150: Raptor +0.31 and Platinum +0.15 above the

model's average, while the work-truck XL is −0.17; the Explorer ST adds 0.15 and the Ford Escape as a model is −0.14.

Depreciation rates

A log-linear model, log(ratio) = a + b × age + c × miles/10,000 + controls for source, condition, title, seller and model, turns its coefficients into percentages: 1 − e^b is the share of value lost per extra year with mileage fixed. A second fit leaves mileage out, which gives the total yearly loss including the miles a typical car adds each year.

Group	Listings	Value lost per extra year (mileage fixed)	Value lost per extra 10,000 miles	Value lost per year (age only)
All cars	1,908	**2.8%**	**3.7%**	**4.8%**
Ford	646	3.4%	3.6%	5.2%
Honda	633	3.0%	3.6%	5.4%
Toyota	629	2.3%	3.9%	4.1%
Ford Explorer	218	4.4%	4.7%	7.3%
Toyota RAV4	223	4.4%	3.6%	6.3%
Honda Accord	199	4.1%	3.0%	6.1%
Honda CR-V	219	2.8%	4.0%	5.6%
Ford Escape	226	3.3%	4.4%	5.0%
Ford F-150	202	3.0%	2.2%	4.1%
Honda Civic	215	1.6%	4.2%	4.1%
Toyota Camry	205	2.1%	4.2%	3.5%
Toyota Corolla	201	0.7%	3.5%	2.7%

The log model explains 85% of the variance in log retention for all cars (76-91% per model). These rates are measured against each car's own original MSRP in nominal dollars, and recent used-car price inflation props up the ratios of cars a few years old, so they understate the loss buyers would see in today's dollars.

Major drivers of depreciation

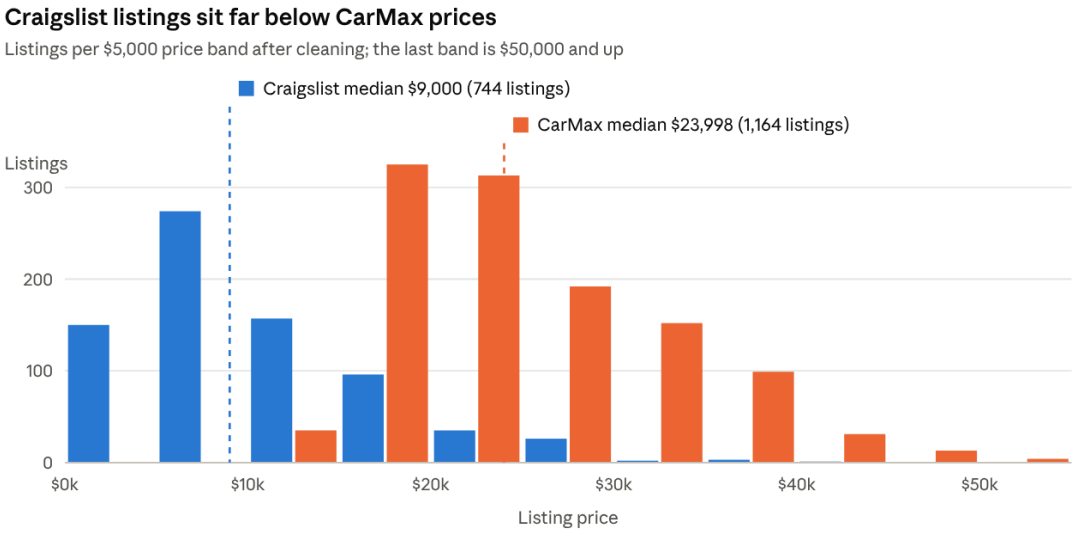
- **Seller channel.** At the same age, CarMax prices sit far above Craigslist prices, and the gap widens with age: median ratio 1.12 vs 0.94 for 0-1-year-old cars, but 0.88 vs 0.37 at 10-11 years. Retail reconditioning, warranty and financing are priced in; private sellers of old cars discount heavily.

- **Mileage.** The learned curve (partial dependence of gradient boosting) drops from 0.93 at 10,000 miles to 0.85 at 50,000, 0.70 at 100,000 and 0.57 at 200,000 miles. The loss is steepest between about 20,000 and 100,000 miles and flattens past 150,000.
- **Age.** With mileage and source held at their observed values, the learned age curve is flat for the first 8 years (0.86 to 0.81), then steps down to 0.73 by year 10 and 0.63 by year 19, when cars fall out of warranty and into the private-sale market.
- **Brand and model.** Median retention at 9-11 years is 0.87 for Toyota and 0.81 for Honda but 0.49 for Ford; at 12-14 years it is 0.53, 0.47 and 0.30. The Corolla and Camry lose the least per year and the Explorer and RAV4 the most (the RAV4 starts from a very high ratio when new).
- **Trim and equipment.** High trims (Raptor, Platinum, XSE, ST) and listings that mention more premium features keep more of the base MSRP; base and work trims keep less.
- **Condition signals.** Title status and "needs work" text matter for the few cars that have them: rebuilt titles have a median ratio of 0.68 and salvage titles 0.38, against 0.84 for clean titles.

8. Visualizations

The three charts below cover the required views: the data's characteristics, the feature ranking and the model comparison. The repository's `figures/` folder holds eleven more detailed figures, listed at the end of this section.

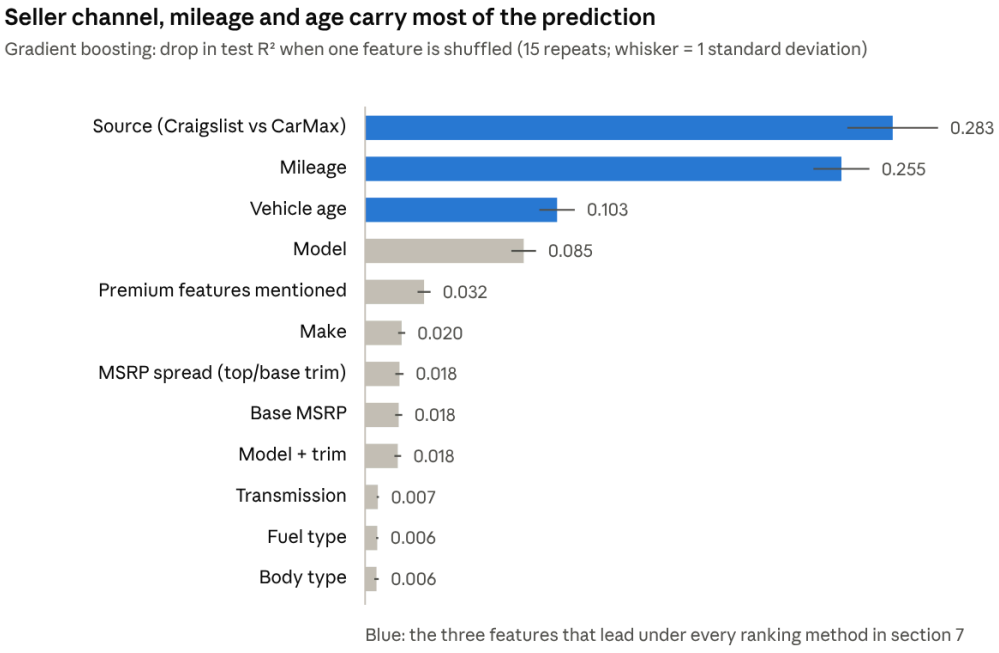
Data characteristics



data/processed/model_ready.csv · 1,908 listings after cleaning

The two sources barely overlap in price. Craigslist is dominated by cars under $15,000, while CarMax rarely lists anything under $15,000. This is why "source" is the strongest feature: it separates a retail market from a private-sale market, and the models need it to read the same age and mileage correctly in both.

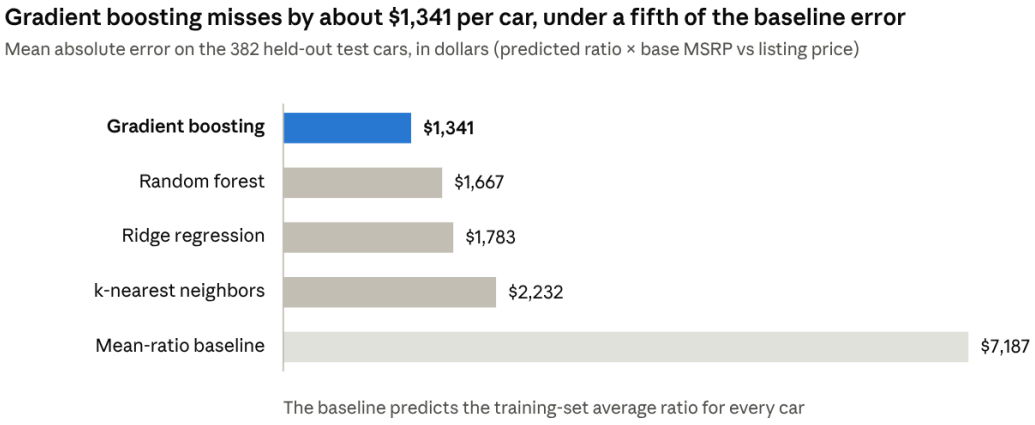
Feature ranking



results/permutation_importance.csv · gradient boosting, 382 test cars

Shuffling the source, mileage or age column costs the model 0.10-0.28 of test R^2 each, far more than any other feature. Model identity and equipment come next; the trim and the two MSRP features add smaller amounts because part of their information is already in the model name.

Model comparison



results/model_comparison.csv · test set, 382 cars

All four models cut the baseline's error by at least two thirds. The ordering in dollars matches the ratio metrics in section 6: gradient boosting first, then the random forest and Ridge regression close together, and k-nearest neighbors last.

Figures in the repository

File	What it shows	What it tells us
fig01_data_overview.png	Price, mileage, age and retention-ratio histograms, Craigslist vs CarMax	the two sources cover different parts of the market
fig02_make_model_counts.png	Listings per make and model, stacked by source	every model has 199-226 listings, so no model dominates
fig03_retention_vs_age_mileage.png	Retention vs age (median line per make) and vs mileage (median per source)	the ratio falls steadily with both; Ford falls fastest
fig04_correlation_heatmap.png	Spearman correlations between numeric attributes	age and mileage are strongly correlated (0.74), which explains the split credit between them
fig05_model_comparison.png	CV and test MAE, RMSE and R^2 for the four models, with the baseline	CV and test scores agree, so the models generalize
fig06_predicted_vs_actual.png	Gradient-boosting predictions vs actual ratios on the test set	errors are larger for Craigslist cars and for ratios above 1.2
fig07_feature_importance.png	Permutation importance of the top 15 features, with error bars	same ranking as the chart above, in more detail
fig08_importance_all_models.png	Importance of the top 10 features in each of the four models	the leading features hold across models; Ridge leans more on age
fig09_ridge_coefficients.png	Largest Ridge coefficients, colored by sign	direction of effects, led by F-150 trims, age and mileage
fig10_depreciation_curves.png	Partial dependence on age and mileage, and on age by make	the learned depreciation curves of section 7
fig11_depreciation_rates.png	% of value lost per year and per 10,000 miles, by model	Explorer and RAV4 lose fastest, Corolla slowest

9. Conclusion

Summary of findings

- **Prediction.** Gradient boosting predicts the retention ratio with test R^2 0.935 and MAE 0.055 (about $1,341 per car; half the test cars are within 4.6% of their listing price). Ridge regression and the random forest reach about 0.91, k-nearest neighbors 0.87, and all far beat the mean-ratio baseline.
- **Most important factors.** Where the car is sold (retail CarMax vs private-heavy Craigslist), its mileage and its age lead under every ranking method, followed by the model, the equipment mentioned in the listing, the make and the trim.
- **Major drivers of depreciation.** With everything else fixed, a car loses about 3.7% of its base-MSRP value per extra 10,000 miles and 2.8% per extra year; counting the miles a typical car adds each year, the loss is about 4.8% per year. Toyota and Honda hold value far longer than Ford (median ratio at 9-11 years: 0.87 and 0.81 vs 0.49), and the gap between retail and private-sale prices widens as cars age. Salvage and rebuilt titles and "needs work" descriptions mark clear discounts.
- **Text features.** Description flags such as "one owner" or "needs work" match real price differences inside each source, but add little to the model because mileage, age and source already carry most of the same information.

What I learned

- Most of the effort in a multi-source project goes into integration: the same fact (year, drivetrain, price) arrives in different names, formats and levels of reliability, and each needs its own rule. Decoding the model year from the VIN caught sellers' typos that no single field would reveal.
- Scraping depends on how each site behaves. Server-rendered pages, a JSON endpoint behind a search page and rate limits each needed a different approach, and polite request spacing matters.
- Encodings should be tested rather than assumed. Target encoding is the textbook answer for high-cardinality features, but pooled one-hot encoding was better on this data.
- Correlated features split their importance differently in each model (age vs mileage), so a ranking from one model alone can mislead. Comparing several methods gives a more honest picture.
- Logging every cleaning rule with its counts made it easy to explain where each listing went.

Challenges

- Two candidate sources were not usable without getting past a human-verification page or a login (Cars.com, Facebook Marketplace), so CarMax became the second source.
- U.S. News rate-limited the MSRP collector before the Nissan Rogue and Sentra pages were read, which cut the study from 4 makes to 3.
- Craigslist data is noisy: placeholder and down-payment prices, years that disagree with the title or VIN, missing VINs (56%) and trims missing from 32% of titles.

- The two sources cover different market segments, so the source feature is tangled with age, mileage and price level.

Limitations

- **The MSRP is the base trim's.** U.S. News gives a range per model year, not each car's trim price, so high-trim cars can show ratios above 1. Trim-level MSRP would give a cleaner target.
- **Asking prices, not sale prices.** Listings show what sellers hope to get; final sale prices are usually lower, especially on Craigslist.
- **One day, one region.** The data is a snapshot from October 23, 2026 (Craigslist within 150 miles of Philadelphia), taken while used-car prices are still above pre-2021 levels, and the ratios use MSRPs in nominal dollars.
- **Narrow scope.** Three makes and nine popular models, model years 2007-2027; the results may not hold for luxury, electric or rare vehicles.
- **Self-reported and missing attributes.** Condition is the seller's opinion and is absent for every CarMax car; description flags come from keyword rules that can miss negations or slang.
- **Model-based curves are approximate.** Partial dependence assumes features can vary independently, so the age curve partly extrapolates CarMax cars to ages CarMax does not sell.

10. AI usage statement

Tool used: Claude (Anthropic), in the Claude desktop app, including its built-in browser and its access to a project folder on my computer. It was used throughout the project

What Claude did

- Proposed the design: the two listing sources, the MSRP source, the makes and models, the feature list and the four algorithms.
- Wrote the code in this repository: the three JavaScript collectors and the Python pipeline (`parse_raw.py`, `integrate.py`, `features.py`, `train.py`, `analysis.py`, `visualize.py`, `run_pipeline.py`).
- Ran the collectors in the Claude app's browser on my computer and ran the pipeline on my machine; chose the cleaning rules, the encoding experiment and the hyper-parameter grids; made the figures.

What I did

- Chose this project and set its requirements from the assignment.
- Supervised the data collection on my own computer: approved each website the scrapers were allowed to visit, approved saving the data files to my computer and gave access to the project folder.
- read through and tested the code, re-ran `run_pipeline.py` and checked that the numbers in this report match `results/`, checked a sample of listings against the websites, rewrote

sections in my own words, made changes to the analysis and code for parts that were incomplete or where I asked for templates.

- Drafted this report, including the discussion sections.